Лабораторна робота №4

«Logical data: operators and expressions»

«Printing data»

«Near zero float numbers»

«Getting input from user»

група ІСТ-21

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Варіант: 18

Lab 1.6.1

```cpp
#include <iostream>
using namespace std;

int main(void) {
    bool answer;
    int value;

    cout << "Enter a value: ";
    cin >> value;

    answer = (value >= 0 && value < 10) ||
             (value * 2 < 20 && value - 2 > -2) ||
             (value - 1 > 1 && value / 2 < 10) ||
             (value == 111);

    cout << (answer ? "THAT'S TRUE :)" : "THAT'S NOT TRUE :(") << endl;
    return 0;
}
```

```
PS D:\C++\Lab4> .\lab1_6_1
Enter a value: 4
THAT'S TRUE :)
```

```
Enter a value: 100
THAT'S NOT TRUE :(
```

Lab 1.7.1

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    double v1, v2, v3, v4, v5;

    cin >> v1 >> v2 >> v3 >> v4 >> v5;

    cout << v1 << endl;
    cout << fixed << setprecision(2) << v2 << endl;
    cout << fixed << setprecision(6) << v3 << endl;
    cout << fixed << setprecision(2) << v4 << endl;
    cout << fixed << setprecision(0) << v5 << endl;

    return 0;
}
```

```
PS D:\C++\Lab4> .\lab1_7_1
2.3 2.3 2.123456 2.123456 2.123456
2.3
2.30
2.123456
2.12
2
PS D:\C++\Lab4>
```

Lab 1.7.2

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int num1, num2;
    cin >> num1 >> num2;

    double res1 = 1.0 / num1;
    double res2 = 1.0 / num2;

    double epsilon = 0.000001;

    if (fabs(res1 - res2) < epsilon) {
        cout << "Results are equal (by 0.000001 epsilon)" << endl;
    } else {
        cout << "Results are not equal (by 0.000001 epsilon)" << endl;
    }

    return 0;
}
```

```
PS D:\C++\Lab4> .\lab1_7_2
1 1
Results are equal (by 0.000001 epsilon)
```

```
1011 1112
Results are not equal (by 0.000001 epsilon)
```

Lab 1.7.3

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b, c, d;
    cin >> a >> b >> c >> d;

    if ((a >= 1 && a <= 255) &&
        (b >= 1 && b <= 255) &&
        (c >= 1 && c <= 255) &&
        (d >= 1 && d <= 255)) {

        cout << a << "." << b << "." << c << "." << d << endl;
    } else {
        cout << "Error: IP components must be between 1 and 255." << endl;
    }

    return 0;
}
```

```
PS D:\C++\Lab4> .\lab1_7_3
4 2 225 212
4.2.225.212
```